MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, LOW-POWER
TYPE 2N466

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a PNP, germanium, low-power transistor.

1.2 Physical dimensions. See figure 1 (TO-5).

* 1.3 Maximum ratings.

P_T 1/	V_{CBO}	V_{CER}	V_{EBO}	T_{stg} and T_{op}
mW	Vdc	Vdc	Vdc	°C
150	-35	-25	-12	-55 to +100

1/ Derate linearly 2 mW/°C for $T_A > 25$°C.

1.4 Primary electrical characteristics.

Limits	h_{fe} V_{CB} = -6 Vdc I_E = 1 mAdc	f_{hfb} V_{CB} = -6 Vdc I_E = 1 mAdc	NF V_{CB} = -2.5 Vdc I_E = 0.5 mAdc R_g = 100 ohms	C_{obo} V_{CB} = -6 Vdc I_E = 0 100 kHz $\leq f \leq$ 1 MHz
		MHz	dB	pF
Min	54	0.5	---	---
Max	130	---	22	60

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

* MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. Transistors shall be of the design, construction, and physical dimensions shown on figure 1.

* 3.3.1 Lead material and finish. Lead material shall be Kovar or alloy 52. Lead finish shall be gold-plated. (Leads may be tin-coated if specified in the contract or order, and this requirement shall not be construed as adversely affecting the qualified-product status of the device, or applicable JAN marking (see 6.2)).

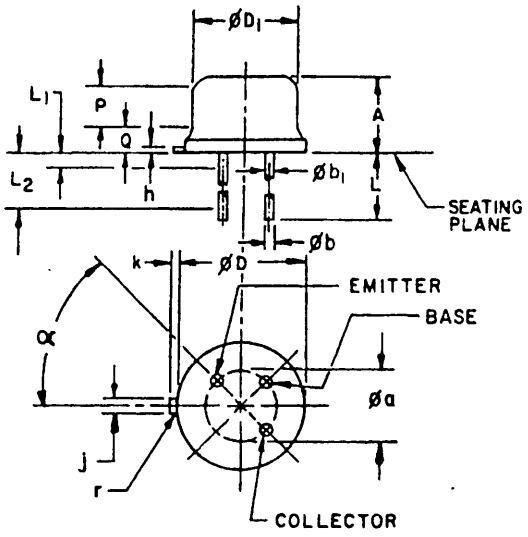
* 3.3.1.1 Selectivity of lead material. Where choice of lead material (see 3.3.1 above) is desired, it shall be specified in the contract or order (see 6.2).

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

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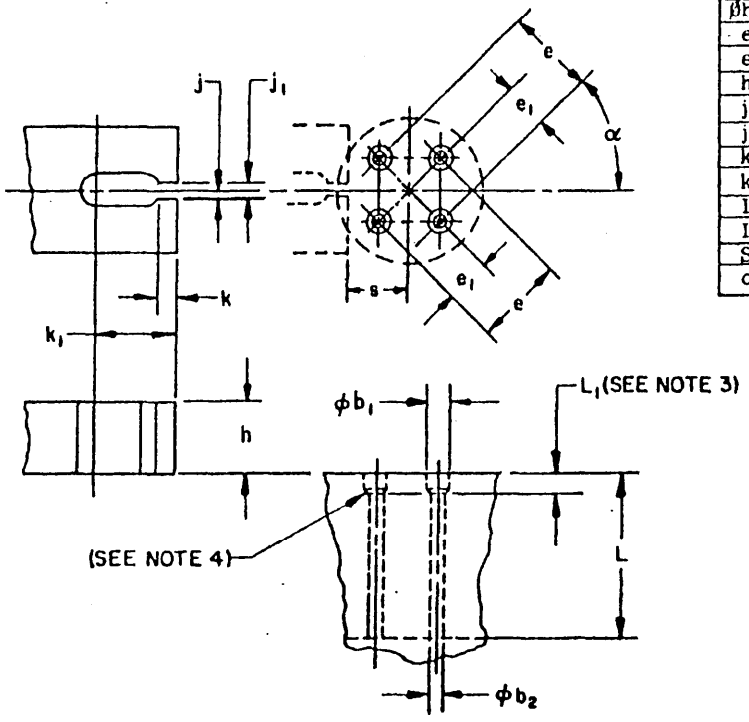


LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.240	.260	6.10	6.60	
Øa	.200 TP		5.08 TP		6
Øb	.016	.021	.41	.53	7, 8
$Øb_1$	.016	.019	.41	.48	7, 8
ØD	.335	.370	8.51	9.40	
$ØD_1$	.305	.335	7.75	8.51	
h	.009	.041	.23	1.04	
j	.028	.034	.71	.86	2
k	.029	.045	.74	1.14	3
L	1.500	1.750	38.10	44.45	7, 8
L_1	---	.050	---	1.27	7, 8
L_2	.250	---	6.35	---	7, 8
P	.100	---	2.54	---	5
Q	---	.050	---	1.27	4
r	---	.010	---	.25	10
α	45° TP		45° TP		6

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Beyond r(radius) maximum, j shall be held for a minimum length of .011(.28 mm).
3. k measured from maximum ØD.
4. Outline in this zone is not controlled.
5. $ØD_1$ shall not vary more than .010(.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gage plane .054 + .001-.000(1.37+.03-.00 mm) below seating plane shall be within .007(.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gage and gaging procedure shown in figure 2.
7. $Øb_1$ applies between L_1 and L_2. Øb applies between L_2 and L minimum. Diameter is uncontrolled in L_1 and beyond L minimum.
8. All three leads.
9. All leads electrically insulated from the case.
10. r(radius) applies to both inside corners of tab.

* FIGURE 1. Physical dimensions of transistor type 2N466 (TO-5).

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LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
ϕb_1	.0595	.0605	1.51	1.54	
ϕb_2	.0325	.0335	.83	.85	
e	.1995	.2005	.07	5.09	
e_1	.0995	.1005	2.53	2.55	
h	.150 Nominal		3.81 Nominal		
j	.0175	.0185	.44	.46	
j_1	.0350	.0355	.89	.90	
k	.009	.011	.23	.28	
k_1	.125 Nominal		3.18 Nominal		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
S	.182	.199	4.62	5.05	1
α	44.90°	45.10°	44.90°	45.10°	

NOTES:
1. The location of the tab locator within the limits indicated will be determined by the tab and flange dimensions of the device being checked.
2. The following gaging procedure shall be used:

   The device being measured shall be inserted until its seating plane is .125(3.18 mm) ±.010(.25 mm) from the seating surface of the gage. A force of 8 ± .5 oz shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gage.

   The use of a pin straightener prior to insertion in the gage is permissible.

3. Gaging plane.
4. Drill angle.

FIGURE 2. Gage for lead and tab location for transistor type 2N466.

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TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Breakdown voltage, collector to base	3001	Bias condition D: I_C = -10 μAdc		BV_{CBO}	-35	---	Vdc
Breakdown voltage, collector to emitter	3011	Bias condition B; I_C = -600 μAdc: R_{BE} = 10 kilohms		BV_{CER}	-25	---	Vdc
Emitter to base cutoff current	3061	Bias condition D: V_{EB} = -12 Vdc		I_{EBO}	---	-12	μAdc
Collector to base cutoff current	3036	Bias condition D: V_{CB} = -35 Vdc		I_{CBO}	---	-20	μAdc
Subgroup 3			15				
Small-signal short-circuit forward-current transfer ratio	3206	V_{CB} = -6 Vdc: I_E = 1 mAdc		h_{fe}	54	130	---
Small-signal short-circuit forward-current transfer ratio cutoff frequency	3301	V_{CB} = -6 Vdc: I_E = 1 mAdc		f_{hfb}	0.5	---	MHz
Open circuit output capacitance	3236	V_{CB} = -6 Vdc; I_E = 0: 100 kHz ≤ f ≤ 1 MHz		C_{obo}	---	60	pF
Small-signal short-circuit input impedance	3201	V_{CB} = -6 Vdc; I_E = 1 mAdc		h_{ib}	18	45	ohms
Small-signal open-circuit output admittance	3216	V_{CB} = -6 Vdc; I_E = 1 mAdc		h_{ob}	---	1	μmho
Noise figure	3246	V_{CB} = -2.5 Vdc: I_E = 0.5 mAdc: R_L = 100 kilohms: R_g = 100 ohms		NF	---	22	dB
Subgroup 4			15				
High-temperature operation:		T_A = +71°C					
Collector to base cutoff current	3036	Bias condition D: V_{CB} = -12 Vdc		I_{CBO}	---	-100	μAdc
Small-signal short-circuit forward-current transfer ratio	3206	V_{CB} = -6 Vdc; I_E = 1 mAdc		h_{fe}	---	250	---
Low-temperature operation:		T_A = -55°C					
Small-signal short-circuit forward-current transfer ratio	3206	V_{CB} = -6 Vdc: I_E = 1 mAdc		h_{fe}	25	---	---

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TABLE II. Group B inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			20				
Physical dimensions	2066	See figure 1		---	---	---	---
* Subgroup 2			15				
Solderability	2026			---	---	---	---
Thermal shock (temperature cycling)	1051	Test condition A, except in step 3; $T_A = +100° \, ^{+5°}_{-0°}C$		---	---	---	---
Thermal shock (glass strain)	1056	Test condition A		---	---	---	---
Hermetic seal	1071	Test condition G or H for fine leaks; test condition A, C, D, or F for gross leaks		---	---	$1x10^{-7}$	atm cc/s
Moisture resistance	1021			---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias condition D; V_{CB} = -35 Vdc		I_{CBO}	---	-20	μAdc
Small-signal short-circuit forward-current transfer ratio	3206	V_{CB} = -6 Vdc; I_E = 1 mAdc		h_{fe}	54	130	---
* Subgroup 3			15				
Shock	2016	Nonoperating; 1,500 G, for 0.5 ms, 5 blows in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
Vibration, variable frequency	2056			---	---	---	---
Constant acceleration	2006	20,000 G; in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
End points: (Same as subgroup 2)							
* Subgroup 4			20				
Terminal strength (lead fatigue)	2036	Test condition E		---	---	---	---
End points:							
Hermetic seal	1071	Test condition G or H for fine leaks; test condition A, C, D, or F for gross leaks		---	---	$1x10^{-7}$	atm cc/s

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TABLE II. Group B inspection -Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
* Subgroup 5			20				
Salt atmosphere (corrosion)	1041			---	---	---	---
* Subgroup 6			10				
High-temperature life (nonoperating)	1032	T_{stg} = +100°C (see 4.3.4)		---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias condition D; V_{CB} = -35 Vdc		I_{CBO}	---	-25	μAdc
Small-signal short-circuit forward-current transfer ratio	3206	V_{CB} = -6 Vdc; I_E = 1 mAdc		h_{fe}	44	150	---
* Subgroup 7			10				
Steady-state operation life	1027	P_T = 150 mW; V_{CB} = -35 Vdc (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6)							

TABLE III. Group C inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
* Subgroup 1			10				
Resistance to solvents	---	MIL-STD-202, Method 215 (see 4.4.1)		---	---	---	---
* Subgroup 2			λ = 10				
High-temperature life (nonoperating)	1031	T_{stg} = +100°C (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6 of group B)							
* Subgroup 3			λ = 10				
Steady-state operation life	1026	P_T = 150 mW; V_{CB} = -35 Vdc (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6 of group B)							

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of group A, B, and C inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 Group C inspection. Group C inspection shall consist of the tests specified in table III. This inspection shall be conducted on the initial lot and thereafter every 6 months during production.

* 4.3.4 Group B and group C life-test samples. Samples that have been subjected to group B, 340-hour life-test, may be continued on test to 1,000 hours in order to satisfy group C life-test requirements. These samples shall be predesignated, and shall remain subjected to the group C, 1,000-hour acceptance evaluation after they have passed the group B, 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1,000 hours shall be computed for 1,000-hour acceptance criteria, see 4.3.3.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III, and as follows:

* 4.4.1 Resistance to solvents. Transistors shall be subjected to tests in accordance with method 215 of MIL-STD-202. The following details shall apply:

(a) All areas of the transistor body where marking has been applied shall be brushed.
(b) After subjection to the tests, there shall be no evidence of mechanical damage to the device and markings shall have remained legible.

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500, section 5.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

* 6.2 Ordering data.

(a) Lead finish if other than gold-plated (see 3.3.1).
(b) Selectivity of lead material (see 3.3.1.1).

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - EC
Air Force - 17

Review activities:
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Air Force - 11, 80
DSA - ES

User activities:
Army - AV, SM
Navy - AS, CG, MC, OS, SH
Air Force - 13, 15, 19

Preparing activity:
Army - EL

Agent:
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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-S-19500/51E SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, LOW-POWER TYPE 2n166

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

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